Customer Shopping Behaviour Analysis

1. Project Background

This report studies the buying patterns of customers using a retail dataset of 3,900 transactions. The aim is to understand how much customers spend, which products perform well, and how discounts and subscriptions influence purchases, so that the business can plan its marketing and pricing better. The data was cleaned in Python, queried in PostgreSQL and presented through a Power BI dashboard.

2. Dataset Description

The dataset contains 3,900 records and 18 attributes, which can be grouped as follows:

- **Customer profile:** age, gender, location and subscription status
- **Transaction details:** item purchased, category, amount spent, season, size and color
- **Buying behaviour:** discount usage, promo code usage, previous purchases, purchase frequency, review rating and shipping type

The only column with missing entries was Review Rating, where 37 values were blank. Across all records, the average spend per purchase is $59.76 and the average review rating is 3.75 out of 5.

3. Data Preparation in Python

1. Imported the file using pandas and inspected its structure and summary statistics with `df.info()` and `df.describe()`.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	Na
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	Na
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	Na
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	Na
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	Na
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	Na
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	Na

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

2. Handled the 37 blank Review Rating entries by filling each one with the median rating of its own product category.
3. Converted all column names to **snake case** for consistency and easier querying in SQL.
4. Added two derived fields**:** **`age_group`**, created by bucketing customer ages, and **`purchase_frequency_days`**, built from the purchase frequency data.
5. Compared `discount_applied` with `promo_code_used`. Since both carried the same information, `promo_code_used` was removed.
6. Loaded the cleaned DataFrame into a PostgreSQL database for the SQL stage.

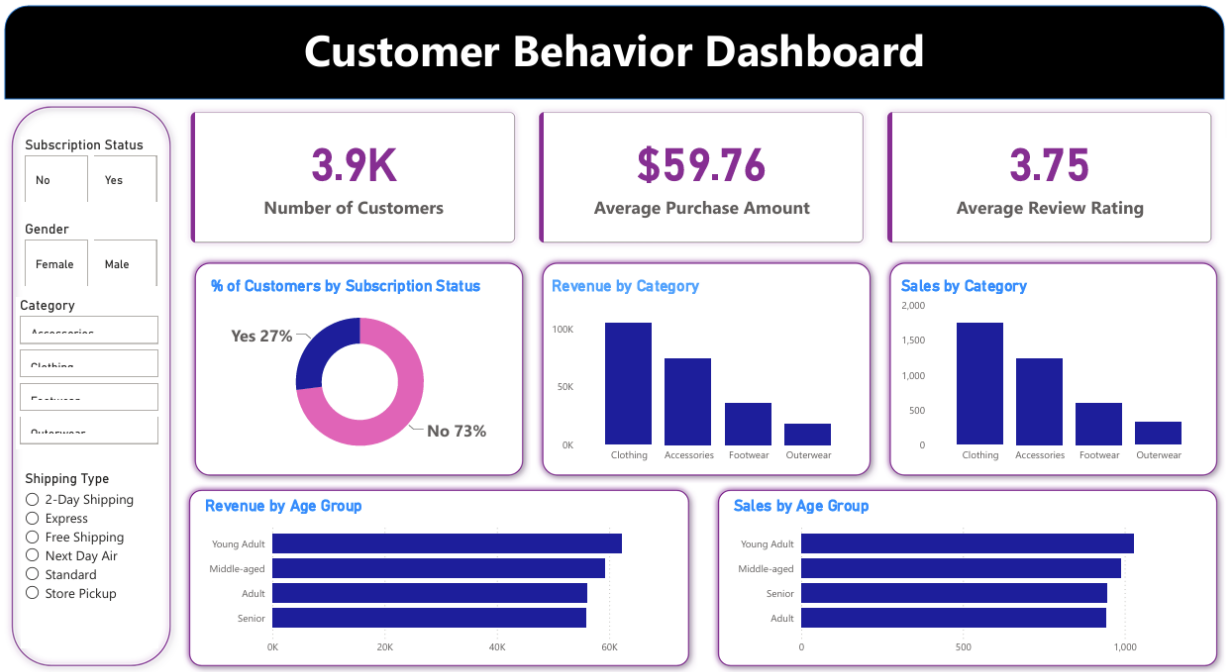
4. Business Analysis in SQL

#	Business Question	Finding
1	Which gender brings in more revenue?	Male customers: $157,890; Female customers: $75,191
2	Which customers used a discount but still spent above the average?	839 customers
3	Which products have the best average rating?	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)
4	Does shipping type affect average spend?	Express: $60.48; Standard: $58.46
5	How do subscribers compare with non-subscribers?	Subscribers: 1,053 customers, average spend $59.49, revenue $62,645. Non-subscribers: 2,847 customers, average spend $59.87, revenue $170,436
6	Which products depend most on discounts?	Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%)

7	How are customers split across segments?	Loyal: 3,116; Returning: 701; New: 83
8	What are the most purchased products in each category?	Jewelry (Accessories), Blouse (Clothing), Sandals (Footwear), Jacket (Outerwear)
9	Are customers with more than 5 purchases more likely to subscribe?	Non-subscribers: 2,518; Subscribers: 958
10	Which age group contributes the most revenue?	Young Adult: $62,143; Middle-aged: $59,197; Adult: $55,978; Senior: $55,763

5. Power BI Dashboard

An interactive dashboard was created in Power BI to present the results visually. It allows filtering by subscription status, gender, category and shipping type. The top cards show 3.9K customers, an average purchase of $59.76 and an average rating of 3.75. The charts cover the subscription split (27% subscribed, 73% not subscribed), revenue and sales by category, where Clothing leads, and revenue and sales by age group.



6. Key Insights

- **Gender:** Male customers contribute most of the revenue mainly because they make up about 68% of the customer base (2,652 of 3,900), not necessarily because each of them spends more.
- **Subscriptions:** Subscribers spend about the same per order as non-subscribers ($59.49 vs $59.87). Only 27% of customers are subscribed, so there is plenty of room to grow.
- **Shipping:** Express shipping customers spend roughly $2 more per order, which is a small difference.

- **Customer segments:** About 80% of customers fall in the Loyal segment and only 83 are New, which suggests the business is not attracting many new buyers.
- **Categories:** Clothing leads in both sales and revenue, followed by Accessories, Footwear and Outerwear.
- **Age groups:** Revenue is spread fairly evenly. Young Adults are at the top, but Seniors are not far behind.
- **Discounts:** Hat, Sneakers and Coat are bought on discount in roughly half of their orders.

7. Recommendations

1. **Grow subscriptions** by introducing exclusive perks for subscribers, since only 27% of customers are subscribed.
2. **Launch a loyalty program** that rewards repeat purchases and moves Returning customers toward the Loyal tier.
3. **Revisit the discount policy** so that sales boosts do not eat into margins, especially for products that rely heavily on offers.
4. **Feature top-rated and best-selling products** in marketing campaigns.
5. **Target the right audience** by focusing on Young Adult and Middle-aged customers, who bring in the most revenue, and on express-shipping users.

8. Conclusion

The analysis shows that the business has a solid base of repeat customers and that Clothing is its biggest revenue source. Subscriptions and discounting are the two areas that need the most attention. The findings and the dashboard can support the business in planning marketing and pricing decisions.